Establish sequential convergence of Lloyd's algorithms without globally subanalytic densities

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Website #1 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Fix $N \geq 1$. Let μ be a compactly supported absolutely continuous probability measure on $\mathbb{R}^d$, $d\mu(x) = f(x)\,dx$. For $Y = (y_1, \dots, y_N) \in (\mathbb{R}^d)^N$, define

$$G_N(Y) = \frac{1}{2}\int_{\mathbb{R}^d} \min_{1\leq i\leq N} \|x - y_i\|^2 \, d\mu(x), \qquad F_N(Y) = \frac{1}{2} W_2^2\Big(\mu, \frac{1}{N}\sum_{i=1}^{N} \delta_{y_i}\Big).$$

For optimal quantization, Lloyd's map is the Voronoi-barycenter map T_N, and the iterates are $Y_{n+1} = T_N(Y_n)$. For uniform quantization, the analogous semi-discrete OT/Laguerre-barycenter map is B_N, and the iterates are $Y_{n+1} = B_N(Y_n)$.

The question asks for conditions on μ, beyond the globally subanalytic / o-minimal definability assumptions used by Portales–Cazelles–Pauwels, that guarantee *sequential convergence* of the whole sequence $Y_n \to Y_\infty$. I interpret the strongest natural reading: conditions guaranteeing convergence of the labeled vector Y_n (not merely of the associated discrete measure modulo permutation), for every admissible initialization at the fixed level N.

Alternative readings considered

Two ambiguities matter.

- *Modulo permutation vs. labeled vectors.* Since the measures $\sum_i \pi_i \delta_{y_i}$ and $\frac{1}{N}\sum_i \delta_{y_i}$ are permutation-invariant but the problem asks for $Y_n \to Y_\infty$, I work with labeled vectors. In the one-dimensional uniform theorem below I make the sorting permutation explicit.
- *Pure regularity hypotheses on f vs. structural hypotheses on the induced objective.* I first aimed at the former, but the strongest rigorous results I could prove in higher dimension are structural: discreteness of the relevant critical set. This is still genuinely a condition on the target measure μ (for the fixed N), because G_N and F_N are induced by μ.

Related literature found

- **Portales–Cazelles–Pauwels (2026),** SOLVES. This is the paper named in the prompt: it proves sequential convergence of both Lloyd schemes under globally subanalytic / analytic density assumptions via the Kurdyka–Łojasiewicz mechanism. I use it only as orientation, not as a proof step. (tse-fr.eu)
- **Emelianenko–Ju–Rand (2008),** TOOL. For classical Lloyd / optimal quantization on a convex bounded domain with density in L^1 positive a.e., they prove weak global convergence: every accumulation point is a nondegenerate critical point, and moreover $\|Y_{n+1} - Y_n\| \to 0$. This supplies exactly the asymptotic-regularity input needed for a trapping argument. (epubs.siam.org)

- **Du–Emelianenko–Ju (2006),** PARTIAL. They prove global convergence in one dimension for positive smooth densities, and also give a "finite fixed points at each distortion level" criterion for classical Lloyd. They furthermore exhibit a rotationally symmetric disk example with a continuum of critical points, which is a useful red-team test against overstrong convergence claims. (math.cmu.edu)
- **Pagès–Yu (2016),** PARTIAL. They prove pointwise convergence of Lloyd I in higher dimension under a special splitting-type initialization condition. This is strong but initialization-dependent, so it does not answer the "all admissible initializations" reading I adopted. (epubs.siam.org)
- **Mérigot–Santambrogio–Sarrazin (2021),** TOOL. For the uniform semi-discrete problem, they prove that the Lloyd-type energy decreases, the gradients vanish, the whole orbit stays in a compact subset away from the diagonal, and every cluster point is critical. This gives the analogue of weak convergence / asymptotic regularity for B_N. (proceedings.neurips.cc)
- **Bobkov–Ledoux (2019),** TOOL. Their one-dimensional quantile formula

$$W_p^p(\mu,\nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^p \, dt$$

lets one rewrite F_N explicitly on each ordering chamber. This is the key tool behind the one-step convergence theorem for $d = 1$ uniform quantization proved below. (www-users.cse.umn.edu)

Proof sketch

There are two independent mechanisms.

First, for both optimal and uniform quantization, older non-KL papers already provide *weak convergence plus asymptotic regularity*: every accumulation point is critical, and consecutive iterates satisfy $\|Y_{n+1} - Y_n\| \to 0$. Once one has this, a very short compactness argument shows that *discreteness of the critical set* is enough to force full sequential convergence. Intuitively, once the steps are small, the orbit cannot keep jumping between different isolated critical points forever. Thus, beyond o-minimality, a clean sufficient condition is: $\mathrm{Crit}(G_N)$ or $\mathrm{Crit}(F_N)$ is discrete in the admissible configuration space. This recovers the classical finite-critical-set trapping philosophy for ordinary Lloyd, and extends the same logic to the uniform semi-discrete map B_N. (math.cmu.edu)

Second, in one dimension the uniform problem is far more rigid than the general theory suggests. After sorting the support points, the quantile representation of W_2 shows that F_N splits into a sum of N independent quadratic functions, one on each mass block $((i-1)/N, i/N]$. Therefore the Lloyd update does not depend on the current locations at all, except through their order: it sends the i-th ordered point to the barycenter of the i-th quantile block of μ. Hence B_N is piecewise constant on each ordering chamber, and the iteration stabilizes after one step. (www-users.cse.umn.edu)

Main result

Theorem 1. *Let $N \geq 1$.*

1. ***Optimal quantization.*** *Assume that μ has compact convex support $\Omega \subset \mathbb{R}^d$ and density $f \in L^1(\Omega)$ which is positive almost everywhere on Ω. Set*

$$\mathrm{Crit}(G_N) := \{Y \in \Omega^N \setminus D_N : \nabla G_N(Y) = 0\}.$$

If $\mathrm{Crit}(G_N)$ *is discrete in* $\Omega^N \setminus D_N$*, then for every* $Y_0 \in \Omega^N \setminus D_N$*, the Lloyd iterates*

$$Y_{n+1} = T_N(Y_n)$$

converge to a single point $Y_\infty \in \mathrm{Crit}(G_N)$*.*

2. ***Uniform quantization.*** *Assume that* μ *is absolutely continuous and supported on a compact convex set. Set*

$$\mathrm{Crit}(F_N) := \{Y \in (\mathbb{R}^d)^N \setminus D_N : \nabla F_N(Y) = 0\}.$$

If $\mathrm{Crit}(F_N)$ *is discrete in* $(\mathbb{R}^d)^N \setminus D_N$*, then for every* $Y_0 \in (\mathbb{R}^d)^N \setminus D_N$*, the Lloyd-type iterates*

$$Y_{n+1} = B_N(Y_n)$$

converge to a single point $Y_\infty \in \mathrm{Crit}(F_N)$*.*

3. ***Explicit one-dimensional uniform theorem.*** *Assume* $d = 1$*, and let* μ *be any compactly supported absolutely continuous probability measure on* $\mathbb{R}$*. Let* $Q := F_\mu^{-1}$ *be its quantile function. For any* $Y = (y_1, \dots, y_N) \in \mathbb{R}^N \setminus D_N$*, let* σ *be the permutation such that*

$$y_{\sigma(1)} < \cdots < y_{\sigma(N)}.$$

Then

$$(B_N(Y))_{\sigma(i)} = N \int_{(i-1)/N}^{i/N} Q(u)\, du, \qquad i = 1, \dots, N.$$

In particular, B_N *is constant on each ordering chamber, and*

$$B_N(B_N(Y)) = B_N(Y)$$

for every $Y \in \mathbb{R}^N \setminus D_N$*. So the one-dimensional uniform Lloyd iteration converges after a single step.*

Proof

Lemma 1. *Let* $K \subset \mathbb{R}^m$ *be compact, and let* $(x_n)_{n \geq 0} \subset K$*. Assume that*

$$\|x_{n+1} - x_n\| \longrightarrow 0$$

and that the accumulation set

$$\omega(x) := \left\{ z \in K : \exists n_k \uparrow \infty,\ x_{n_k} \to z \right\}$$

is discrete in K*. Then* (x_n) *converges to a single point of* $\omega(x)$*.*

Proof. Because $\omega(x)$ is a compact discrete subset of the metric space K, it is finite. Indeed, if $\omega(x)$ were infinite, compactness would produce an accumulation point of $\omega(x)$ inside K, contradicting discreteness.

Write

$$\omega(x) = \{a_1, \dots, a_r\}.$$

If $r = 1$, say $\omega(x) = \{a_1\}$, then $x_n \to a_1$. For if not, there would exist $\varepsilon > 0$ and a subsequence (x_{n_k}) with $\|x_{n_k} - a_1\| \geq \varepsilon$ for every k. Since K is compact, (x_{n_k}) has a convergent further subsequence, whose limit belongs to $\omega(x)$ but is not a_1, a contradiction.

Assume now $r \geq 2$, and set

$$\eta := \frac{1}{3} \min_{i \neq j} \|a_i - a_j\| > 0.$$

The open balls $B(a_i, \eta)$ are pairwise disjoint. Let

$$U := \bigcup_{i=1}^{r} B(a_i, \eta).$$

I claim that $x_n \in U$ for all sufficiently large n. If not, there would exist infinitely many indices n_k with $x_{n_k} \in K \setminus U$. By compactness of K, a further subsequence converges to some $b \in K \setminus U$. Then $b \in \omega(x)$, contradicting $\omega(x) \subset U$.

Next, choose N_0 such that $\|x_{n+1} - x_n\| < \eta$ for all $n \geq N_0$, and choose $N_1 \geq N_0$ such that $x_n \in U$ for all $n \geq N_1$. For $n \geq N_1$, if $x_n \in B(a_i, \eta)$ and $x_{n+1} \in B(a_j, \eta)$, then

$$\|a_i - a_j\| \leq \|a_i - x_n\| + \|x_n - x_{n+1}\| + \|x_{n+1} - a_j\| < 3\eta,$$

hence $i = j$ by the definition of η. So after time N_1, the index of the ball containing x_n cannot change.

Therefore there exists $i_* \in \{1, \ldots, r\}$ and $N_2 \geq N_1$ such that

$$x_n \in B(a_{i_*}, \eta) \qquad \forall n \geq N_2.$$

Every accumulation point of the tail $(x_n)_{n \geq N_2}$ must then lie in $\overline{B(a_{i_*}, \eta)} \cap \omega(x)$. Since the balls $B(a_i, \eta)$ are pairwise disjoint, this intersection equals $\{a_{i_*}\}$. Hence the tail has a unique accumulation point, and by the $r = 1$ case, $x_n \to a_{i_*}$. □

Proof of Theorem, part (1). Let $\Omega = \mathrm{Supp}(\mu)$. By the theorem of Emelianenko–Ju–Rand quoted in the "Invoked results" section, under the present assumptions (compact convex Ω, density $f \in L^1(\Omega)$ positive a.e.), every accumulation point of the optimal Lloyd orbit is a nondegenerate critical point of the quantization energy, and

$$\|Y_{n+1} - Y_n\| \to 0.$$

Since $Y_n \in \Omega^N$ for all n and Ω^N is compact, the accumulation set $\omega(Y)$ is compact. It is contained in $\mathrm{Crit}(G_N)$, which is discrete by assumption. Hence $\omega(Y)$ is discrete in the compact set Ω^N. The lemma applies, so Y_n converges to a single point $Y_\infty \in \omega(Y) \subset \mathrm{Crit}(G_N)$. (math.cmu.edu)

Proof of Theorem, part (2). For the uniform Lloyd iteration, Mérigot–Santambrogio–Sarrazin prove that the whole orbit (Y_n) stays in a compact subset $K \subset (\mathbb{R}^d)^N \setminus D_N$, and every accumulation point is critical for F_N. They also prove

$$\nabla F_N(Y) = \frac{1}{N}\big(Y - B_N(Y)\big).$$

Since Proposition 2 of their paper gives $\|\nabla F_N(Y_n)\| \to 0$, we get

$$\|Y_{n+1} - Y_n\| = \|B_N(Y_n) - Y_n\| = N \|\nabla F_N(Y_n)\| \longrightarrow 0.$$

Thus the accumulation set $\omega(Y) \subset K$ is compact and contained in the discrete set $\mathrm{Crit}(F_N)$. By the lemma, Y_n converges to a single point $Y_\infty \in \mathrm{Crit}(F_N)$. (proceedings.neurips.cc)

Proof of Theorem, part (3). Fix $Y \in \mathbb{R}^N \setminus D_N$, and let σ sort its coordinates increasingly:

$$y_{\sigma(1)} < \cdots < y_{\sigma(N)}.$$

Let

$$\nu_Y := \frac{1}{N}\sum_{i=1}^{N} \delta_{y_i}.$$

The quantile function of ν_Y is

$$F_{\nu_Y}^{-1}(u) = y_{\sigma(i)} \qquad \text{for } u \in I_i := \Big(\frac{i-1}{N}, \frac{i}{N}\Big].$$

Since μ has compact support, both μ and ν_Y belong to $Z_2(\mathbb{R})$. By the one-dimensional quantile formula for W_2,

$$2F_N(Y) = W_2^2(\mu, \nu_Y) = \int_0^1 \big|Q(u) - F_{\nu_Y}^{-1}(u)\big|^2\, du = \sum_{i=1}^{N} \int_{I_i} \big(Q(u) - y_{\sigma(i)}\big)^2\, du.$$

Define

$$m_i := N \int_{I_i} Q(u)\, du, \qquad i = 1, \dots, N.$$

Differentiating the displayed formula with respect to $y_{\sigma(i)}$ gives

$$\frac{\partial F_N}{\partial y_{\sigma(i)}}(Y) = \int_{I_i} \big(y_{\sigma(i)} - Q(u)\big)\, du = \frac{1}{N}\big(y_{\sigma(i)} - m_i\big).$$

On the other hand, Proposition 1 of Mérigot–Santambrogio–Sarrazin gives

$$\frac{\partial F_N}{\partial y_{\sigma(i)}}(Y) = \frac{1}{N}\Big(y_{\sigma(i)} - (B_N(Y))_{\sigma(i)}\Big).$$

Therefore

$$(B_N(Y))_{\sigma(i)} = m_i \qquad \text{for each } i = 1, \dots, N.$$

So $B_N(Y)$ depends only on the sorting permutation σ, not on the actual values of the y_i.

It remains to show that $m_1 < \cdots < m_N$. For adjacent indices,

$$m_{i+1} - m_i = N \int_{I_{i+1}} Q(v)\, dv - N \int_{I_i} Q(u)\, du = N^2 \int_{I_i}\int_{I_{i+1}} \big(Q(v) - Q(u)\big)\, du\, dv.$$

Because Q is nondecreasing, the integrand is nonnegative almost everywhere, hence $m_{i+1} \geq m_i$. If equality held, then $Q(v) = Q(u)$ for almost every $(u, v) \in I_i \times I_{i+1}$, which forces Q to be almost everywhere constant on $I_i \cup I_{i+1}$. But if $Q \equiv c$ on an interval of positive u-length, then by the defining relation

$$Q(t) = \inf\{x : F_\mu(x) \geq t\},$$

the distribution function F_μ would have a jump at c, i.e. $\mu(\{c\}) > 0$, contradicting absolute continuity of μ. Thus $m_{i+1} > m_i$.

Hence the sorting permutation of $B_N(Y)$ is again σ. Applying the same formula to $B_N(Y)$ itself yields

$$(B_N(B_N(Y)))_{\sigma(i)} = m_i = (B_N(Y))_{\sigma(i)}$$

for every i, so $B_N(B_N(Y)) = B_N(Y)$. This proves one-step convergence.

Orbitwise weakening. In parts (1) and (2), the same argument shows that global discreteness can be replaced by the weaker requirement that the set of critical points met by the compact forward orbit closure be discrete; because the relevant objective values are decreasing, finiteness only on the attained limiting level also suffices. □

Gap to the full statement

What remains open in this manuscript is the genuinely hard part of the prompt: a broad, checkable *higher-dimensional* characterization of measures μ guaranteeing sequential convergence without either (a) KL/definability hypotheses or (b) a direct discreteness assumption on the critical set.

More concretely:

- For optimal quantization in higher dimension, I proved that *discreteness of* $\mathrm{Crit}(G_N)$ is enough, but I did not derive a practical density-side criterion ensuring this discreteness.
- For uniform quantization in higher dimension, I proved the analogous result for $\mathrm{Crit}(F_N)$, but again not a general density-side criterion forcing discreteness.
- For one-dimensional optimal quantization, stronger convergence theorems are known under positive smooth density hypotheses, but I did not rederive them here; my fully explicit new theorem is only for the one-dimensional *uniform* problem. (math.cmu.edu)

Self red-team

- **Counterexample attempt to "weak convergence + compactness already implies full convergence".** It does not. Du–Emelianenko–Ju exhibit a rotationally symmetric example: for the constant density on the unit disk with $N = 2$, the critical points form an entire circle. So some rigidity hypothesis beyond mere compactness/criticality is genuinely necessary. This is exactly why I imposed discreteness of the critical set in parts (1)–(2). (math.cmu.edu)
- **Counterexample attempt to the abstract lemma without asymptotic regularity.** If one only assumes finitely many accumulation points but not $\|x_{n+1} - x_n\| \to 0$, the sequence can alternate between two points forever. So the small-step hypothesis is essential. In the applications it is supplied by Emelianenko–Ju–Rand for T_N, and by the gradient identity plus gradient decay for B_N. (math.cmu.edu)
- **Counterexample attempt to part (3) on disconnected support.** I checked the proof against measures such as the uniform law on $[0, 1] \cup [2, 3]$. The argument survives unchanged, because it only uses the quantile formula and monotonicity of Q; connectedness of the support never enters.
- **Most plausible referee objection #1:** "A compact subset of a discrete set need not be finite." In a metric space it must be finite: any infinite compact subset has an accumulation point, contradicting discreteness. I built this directly into the proof of the lemma.
- **Most plausible referee objection #2:** "In part (3), the power cells might still depend on Y." The proof avoids any appeal to geometric intuition about power cells. Instead it computes F_N exactly on each ordering chamber via the quantile formula, then identifies B_N from the gradient identity. So the one-step formula is derived analytically, not guessed geometrically.
- **Most plausible referee objection #3:** "Why does strict ordering $m_i < m_{i+1}$ hold?" I checked this carefully through the double-integral identity

$$m_{i+1} - m_i = N^2 \int_{I_i} \int_{I_{i+1}} (Q(v) - Q(u))\, du\, dv.$$

Equality would force Q to be a.e. constant on a set of positive u-measure, hence an atom of μ, excluded by absolute continuity.

- **Boundary cases checked.** For $N = 1$, both Lloyd maps collapse in one step to the mean of μ. In part (3), the displayed formula gives exactly $B_1(Y) = \int x\, d\mu(x)$. For optimal quantization, convex support and positivity a.e. are really used through the external weak-convergence theorem; dropping them exits the audited hypothesis range.

Ideas tried

1. **Successful: asymptotic-regularity trapping.** Idea: combine old weak-convergence theorems with a compactness lemma saying "small steps + discrete cluster set $\Rightarrow$ one limit." Outcome: this succeeded and produced parts (1) and (2). Natural next step: replace discreteness by a weaker local KL or Morse–Bott condition near the cluster set.

2. **Successful: one-dimensional quantile decomposition for uniform quantization.** Idea: use the exact 1D quantile formula for W_2 to rewrite F_N on an ordering chamber. Outcome: this succeeded and yielded the stronger statement that B_N is piecewise constant and stabilizes after one step. Natural next step: extend the same argument to arbitrary prescribed weights π, not only the uniform weights $1/N$.

3. **Stalled: direct analyticity beyond o-minimality.** Idea: assume f real-analytic on a neighborhood of the support and prove that G_N or F_N is real-analytic near nondegenerate configurations by differentiating integrals over moving Voronoi / Laguerre cells. Obstruction: the dependence of these integrals on moving polyhedral cuts interacting with a general curved support boundary is delicate; I did not obtain a rigorous analyticity theorem without re-importing heavy external machinery. Natural next step: a careful Reynolds-transport / shape-derivative analysis on strata where the cell combinatorics are fixed.

4. **Stalled: contraction of the 1D optimal Lloyd map.** Idea: compute the Jacobian of the scalar Lloyd map on the ordered simplex and bound its operator norm by < 1 under log-concavity or related monotonicity assumptions on f. Obstruction: I could derive endpoint-derivative formulas for cell barycenters, but not a clean global norm bound valid for all N. Natural next step: revisit the classical 1D contraction literature and try to isolate a matrix monotonicity argument that scales with N.

Invoked results

1. **Emelianenko–Ju–Rand, Theorem 3.8 (weak global convergence).** *Statement used.* If $\Omega \subset \mathbb{R}^d$ is convex and bounded, and $\rho \in L^1(\Omega)$ is positive almost everywhere, then for every initial nondegenerate configuration Z_0, the classical Lloyd iterates Z_i satisfy:

   (a) $\nabla G(Z_i) \to 0$;

   (b) every accumulation point is a nondegenerate critical point of G;

   (c) $\|Z_{i+1} - Z_i\| \to 0$.

   They also explain that the iterates remain in a compact set away from degeneracy. *Classification: TOOL.* Citation: Emelianenko, Ju, Rand, *SIAM J. Numer. Anal.* 46(3), 1423–1441, 2008, DOI 10.1137/070691334. (math.cmu.edu)

Hypothesis audit. In Theorem part (1), $\Omega = \mathrm{Supp}(\mu)$ is assumed compact and convex; since $d\mu = f\,dx$ with $f \in L^1(\Omega)$ positive a.e., Assumption 3.1 of the cited theorem is satisfied exactly; and $Y_0 \in \Omega^N \setminus D_N$ is the required nondegenerate initial configuration. So the theorem applies without modification.

2. **Mérigot–Santambrogio–Sarrazin, Proposition 1 and Proposition 2.** *Statements used.*

   (a) Proposition 1: on $(\mathbb{R}^d)^N \setminus D_N$,

   $$\nabla F_N(Y) = \frac{1}{N}\big(Y - B_N(Y)\big),$$

   and F_N is C^1 there.

   (b) Proposition 2: for the Lloyd-type iterates $Y^{k+1} = B_N(Y^k)$ with $Y^0 \notin D_N$, the energy $F_N(Y^k)$ is decreasing, $\|\nabla F_N(Y^k)\| \to 0$, the whole orbit lies in a compact subset of $(\mathbb{R}^d)^N \setminus D_N$, and every accumulation point is critical.

   Classification: TOOL. Citation: Mérigot, Santambrogio, Sarrazin, *Advances in Neural Information Processing Systems 34*, 2021, arXiv:2106.07911. (proceedings.neurips.cc)

   Hypothesis audit. The cited paper works in the setting where ρ is a probability density on a compact convex domain Ω. In Theorem part (2), I assume exactly that μ is absolutely continuous with compact convex support, and I start from $Y_0 \notin D_N$. Hence both propositions apply.

3. **Bobkov–Ledoux, Theorem 2.10 (1D quantile representation).** *Statement used.* For probability measures $\mu, \nu \in Z_p(\mathbb{R})$ with distribution functions F, G,

   $$W_p^p(\mu,\nu) = \int_0^1 |F^{-1}(t) - G^{-1}(t)|^p\,dt.$$

   Classification: TOOL. Citation: Bobkov, Ledoux, *Memoirs of the AMS* 261(1259), 2019, DOI $10.1090/memo/1259$. (www-users.cse.umn.edu)

   Hypothesis audit. In Theorem part (3), μ has compact support, hence finite second moment, so $\mu \in Z_2(\mathbb{R})$. The discrete measure $\nu_Y = \frac{1}{N}\sum_i \delta_{y_i}$ is finitely supported, hence also in $Z_2(\mathbb{R})$. Therefore the theorem applies with $p = 2$.

Self-confidence

3 — I checked every step of the proof I actually wrote, and the main remaining uncertainty is not a logical gap but whether the structural condition “critical set discrete” is the strongest useful formulation the prompt hoped for.